

Challenge (Habanero)

- Q1.** A recipe for chicken parmesan requires x cups of marinara sauce and y cups of shredded mozzarella. If $x + y \leq 5$ and $2x + y \geq 6$, what is the minimum amount of mozzarella required?
- (A) 1 cup
(B) 2 cups
(C) 3 cups
(D) 4 cups
(E) 5 cups
- Q2.** A construction company is building a house with a rectangular foundation. If the length L and width W must satisfy $L \geq 20$ and $W \leq 30$, what is the maximum area of the foundation?
- (A) 600 sq ft
(B) 800 sq ft
(C) 900 sq ft
(D) 1000 sq ft
(E) 1200 sq ft
- Q3.** A caterer is planning a wedding reception with x tables and y chairs per table. If $x + 2y \leq 50$ and $3x + y \geq 30$, what is the maximum number of tables?
- (A) 10 tables
(B) 12 tables
(C) 15 tables
(D) 18 tables
(E) 20 tables
- Q4.** A chef is making a recipe that requires x cups of flour and y cups of sugar. If $x \geq 2$ and $y \leq 3$, what is the minimum amount of sugar required if $x = 4$?
- (A) 1 cup
(B) 2 cups
(C) 3 cups
(D) 4 cups
- (E) 5 cups
- Q5.** A builder is constructing a staircase with x steps and y inches of railing. If $x + y \leq 100$ and $2x - y \geq 20$, what is the minimum number of steps required?
- (A) 10 steps
(B) 12 steps
(C) 15 steps
(D) 18 steps
(E) 20 steps
- Q6.** A restaurant is serving a dish with x ounces of meat and y ounces of vegetables. If $x \geq 8$ and $y \leq 12$, what is the maximum amount of vegetables that can be served if $x = 10$?
- (A) 8 oz
(B) 10 oz
(C) 12 oz
(D) 14 oz
(E) 16 oz
- Q7.** A contractor is building a fence with x feet of wood and y feet of wire. If $x + 2y \leq 100$ and $3x - y \geq 30$, what is the maximum amount of wood required?
- (A) 30 ft
(B) 40 ft
(C) 50 ft
(D) 60 ft
(E) 70 ft
- Q8.** A bakery is making a cake with x cups of flour and y cups of sugar. If $x \geq 3$ and $y \leq 5$, what is the minimum amount of flour required if $y = 4$?
- (A) 2 cups
(B) 3 cups
(C) 4 cups
(D) 5 cups
(E) 6 cups

Open Ended Questions (FRQ)

- Q9.** A farmer is planting a field with x rows of corn and y rows of soybeans. If $x + 2y \leq 50$ and $3x - y \geq 20$, find the maximum number of rows of corn and the maximum number of rows of soybeans. Show your work and explain your reasoning.
- Q10.** A chef is planning a menu with x dishes and y ingredients per dish. If $x \geq 5$ and $y \leq 10$, find the minimum number of ingredients required if $x = 8$. Explain your reasoning and provide a graph to support your answer.

Chef's Tips

- Tip 1: Encourage students to read the questions carefully and identify the constraints.
- Tip 2: Remind students to use graphing to visualize the feasible region.
- Tip 3: Emphasize the importance of explaining reasoning and showing work.